# STUDENT PERFORMANCE & ACADEMIC ANALYTICS

*Exploratory Data Analysis Report*

## BLEEP EDUCATION

Artificial Intelligence Internship – June Batch

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Project Type: Exploratory Data Analysis (EDA)

## Declaration

I hereby declare that this report presents the Exploratory Data Analysis work completed as part of the BLEEP Education Artificial Intelligence Internship. The report is organized from the completed project notebook and presents the documented dataset understanding, data-quality checks, exploratory analyses, visualizations, findings, conclusions, limitations, and future scope.

## Acknowledgement

I would like to express my appreciation to BLEEP Education for providing the Artificial Intelligence Internship learning experience. The internship provided an opportunity to apply exploratory data analysis techniques to a student-performance dataset and to document the analytical workflow in a structured project. I am grateful for the learning experience and the practical exposure gained through this EDA project.

## Abstract

This project presents an Exploratory Data Analysis of the student-mat dataset with final grade (G3) used as the main indicator of academic performance. The analysis examines the structure and composition of the dataset, numerical and categorical features, data quality, validity and consistency, potential outliers, distributions, group-wise comparisons, scatter relationships, and correlations among numerical variables. The notebook records 395 student observations across 33 columns, with 16 numerical and 17 categorical features. No missing values or duplicate rows were identified. The EDA uses histograms, count plots, bar charts, box plots, scatter plots, and a correlation heatmap to describe patterns associated with final academic performance. The resulting analysis provides a structured view of student demographic, family, social, behavioral, and academic attributes and establishes a foundation for further statistical or predictive analysis.

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# 1. Introduction

## 1.1 Background

Exploratory Data Analysis (EDA) is used to understand the structure, distribution, relationships, and quality of a dataset before deeper statistical or predictive work. In this project, EDA is applied to student-performance data to examine academic outcomes alongside demographic, family, social, behavioral, and study-related attributes.

## 1.2 Project Context

The project was completed as an EDA project during the BLEEP Education Artificial Intelligence Internship – June Batch. The completed notebook provides the analytical source for this report.

## 1.3 Project Motivation

The project is motivated by the need to understand student academic-performance data systematically rather than examining final grades in isolation. Visual and statistical summaries make it possible to identify distributions and associations that can inform later analysis.

## 1.4 Objectives

- Understand the dataset structure and feature types.
- Assess data quality through missing-value, duplicate, validity, and consistency checks.
- Explore individual variable distributions and category frequencies.
- Compare final grades across selected demographic, study, family, and access-related groups.
- Examine relationships between absences, study time, and final grade.
- Identify numerical variables with stronger or weaker linear associations with G3.
- Detect potential outliers using the IQR method.
- Summarize findings and establish a basis for future statistical or predictive analysis.

## 1.5 Scope

The scope is limited to exploratory analysis of the supplied student-mat dataset. The project does not present a predictive model, causal analysis, or time-series analysis.

# 2. Dataset Overview

## 2.1 Dataset Description

The notebook loads student-mat.csv using a semicolon delimiter. The dataset contains student demographic, family, social, behavioral, study, and academic variables.

## 2.2 Dataset Dimensions

The dataset contains 395 student records and 33 variables. The combination of demographic, family, academic, social, and behavioural attributes allows the EDA to examine final performance from several descriptive perspectives without moving into predictive modelling.

| Measure | Value | Source in Notebook |
|---|---|---|
| Rows | 395 | df.shape |
| Columns | 33 | df.shape |
| Numerical features | 16 | select_dtypes |
| Categorical features | 17 | select_dtypes |

## 2.3 Feature Classification

The notebook classifies 16 variables as numerical: age, Medu, Fedu, traveltime, studytime, failures, famrel, freetime, goout, Dalc, Walc, health, absences, G1, G2, and G3. It classifies 17 variables as categorical: school, sex, address, famsize, Pstatus, Mjob, Fjob, reason, guardian, schoolsup, famsup, paid, activities, nursery, higher, internet, and romantic.

## 2.4 Important Variables and Main Academic Performance Variable

The analysis treats G3 as the principal outcome variable for descriptive comparisons. G1 and G2 are earlier-period grades and are also included in the numerical correlation analysis, while variables such as studytime, failures, absences, Medu, Fedu, and internet provide additional context for interpreting patterns in the dataset.

G3 is used as the main indicator of final academic performance. G1 and G2 represent earlier-period grades, while studytime, failures, absences, gender, parental education, and internet access are among the variables examined in the EDA.

## 2.5 Data Dictionary

| Variable | Description |
|---|---|
| school | Student's school identifier/category. |
| sex | Student's gender category. |
| age | Student's age. |
| address | Student residential-address category. |

| Variable | Description |
|---|---|
| famsize | Family-size category. |
| Pstatus | Parent cohabitation/status category. |
| Medu | Mother's education level code. |
| Fedu | Father's education level code. |
| studytime | Weekly study-time category. |
| failures | Number of past class failures. |
| schoolsup | School-support category. |
| famsup | Family-support category. |
| internet | Internet-access-at-home category. |
| absences | Number of school absences. |
| G1 | First-period grade. |
| G2 | Second-period grade. |
| G3 | Final grade / main academic performance indicator. |

# 3. Data Understanding

## 3.1 Number of Rows and Columns

The notebook reports 395 observations and 33 columns. All 395 observations are represented in the data types and descriptive-statistics outputs.

## 3.2 Data Types

The notebook reports 16 int64 numerical columns and 17 string/categorical columns. No column is reported with missing entries in the data-quality check.

## 3.3 Numerical Descriptive Statistics

| Statistic | age | Medu | Fedu | traveltime | studytime | failures | famrel | freetime | goout | Dalc | Walc | health | absences | G1 | G2 | G3 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| count | 395 | 395 | 395 | 395 | 395 | 395 | 395 | 395 | 395 | 395 | 395 | 395 | 395 | 395 | 395 | 395 |
| mean | 16.696 | 2.749 | 2.522 | 1.448 | 2.035 | 0.334 | 3.944 | 3.235 | 3.109 | 1.481 | 2.291 | 3.554 | 5.709 | 10.909 | 10.714 | 10.415 |
| std | 1.276 | 1.095 | 1.088 | 0.698 | 0.839 | 0.744 | 0.897 | 0.999 | 1.113 | 0.891 | 1.287 | 1.390 | 8.003 | 3.319 | 3.762 | 4.581 |
| min | 15 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 3 | 0 | 0 |
| 25% | 16 | 2 | 2 | 1 | 1 | 0 | 4 | 3 | 2 | 1 | 1 | 3 | 0 | 8 | 9 | 8 |
| 50% | 17 | 3 | 2 | 1 | 2 | 0 | 4 | 3 | 3 | 1 | 2 | 4 | 4 | 11 | 11 | 11 |
| 75% | 18 | 4 | 3 | 2 | 2 | 0 | 5 | 4 | 4 | 2 | 3 | 5 | 8 | 13 | 13 | 14 |
| max | 22 | 4 | 4 | 4 | 4 | 3 | 5 | 5 | 5 | 5 | 5 | 5 | 75 | 19 | 19 | 20 |

Values are reproduced from the notebook's df.describe() output and rounded to three decimals where applicable.

## 3.4 Categorical Feature Summary

| Feature | Count | Unique | Top | Frequency |
|---|---|---|---|---|
| school | 395 | 2 | GP | 349 |
| sex | 395 | 2 | F | 208 |
| address | 395 | 2 | U | 307 |
| famsize | 395 | 2 | GT3 | 281 |
| Pstatus | 395 | 2 | T | 354 |
| Mjob | 395 | 5 | other | 141 |
| Fjob | 395 | 5 | other | 217 |

| Feature | Count | Unique | Top | Frequency |
|---|---|---|---|---|
| reason | 395 | 4 | course | 145 |
| guardian | 395 | 3 | mother | 273 |
| schoolsup | 395 | 2 | no | 344 |
| famsup | 395 | 2 | yes | 242 |
| paid | 395 | 2 | no | 214 |
| activities | 395 | 2 | yes | 201 |
| nursery | 395 | 2 | yes | 314 |
| higher | 395 | 2 | yes | 375 |
| internet | 395 | 2 | yes | 329 |
| romantic | 395 | 2 | no | 263 |

The categorical summary is reproduced from the notebook's df.describe(include="object") output.

### 3.5 Initial Observations

The notebook establishes a mixed-type dataset with both numerical and categorical variables. The final grade G3 ranges from 0 to 20, while absences range from 0 to 75. The study-time and failure variables take four and four observed coded categories respectively.

# 4. Data Quality & Cleaning

## 4.1 Missing Value Analysis

The notebook reports 0 total missing values across all 33 columns. No missing-value treatment was therefore required.

## 4.2 Duplicate Analysis

The notebook reports 0 duplicate rows. The conditional cleaning cell therefore made no changes and explicitly records that no duplicate rows were found.

## 4.3 Data Type Verification

The data types were inspected using df.dtypes. The notebook reports 16 int64 numerical fields and 17 string fields.

## 4.4 Validity & Consistency Checks

The notebook also inspects the available categorical values and the ranges of important numerical variables. These checks are intended to confirm that the loaded data are consistent with the recorded feature definitions before visualization and interpretation.

| Variable / Check | Observed range or values | Notebook result |
|---|---|---|
| Age | 15 to 22 | Observed range |
| G1 | 3 to 19 | Observed range |
| G2 | 0 to 19 | Observed range |
| G3 | 0 to 20 | Observed range |
| Absences | 0 to 75 | Observed range |
| School | GP, MS | Two observed categories |
| Gender | F, M | Two observed categories |
| Study time | 1, 2, 3, 4 | Four observed categories |
| Failures | 0, 1, 2, 3 | Four observed categories |

## 4.5 Outlier Detection Using IQR

The notebook applies the interquartile range (IQR) method to age, absences, G1, G2, and G3. A potential outlier is identified when an observation falls beyond the IQR-based lower or upper fence. These observations were identified for review rather than automatically treated as errors.

| Variable | Potential outliers |
|---|---|
| age | 1 |
| absences | 15 |
| G1 | 0 |
| G2 | 13 |
| G3 | 0 |

### 4.6 Cleaning Decisions

The notebook does not report removing or modifying the potential outliers. Since there were no missing values and no duplicate rows, the dataset remained unchanged for the subsequent EDA.

# 5. Exploratory Data Analysis

The EDA is the main analytical section of the project. It combines univariate, bivariate, multivariate/correlation, and outlier-oriented views to describe student performance and related variables. The figures below are the actual visual outputs stored in the completed notebook.

## 5.1 Univariate Analysis

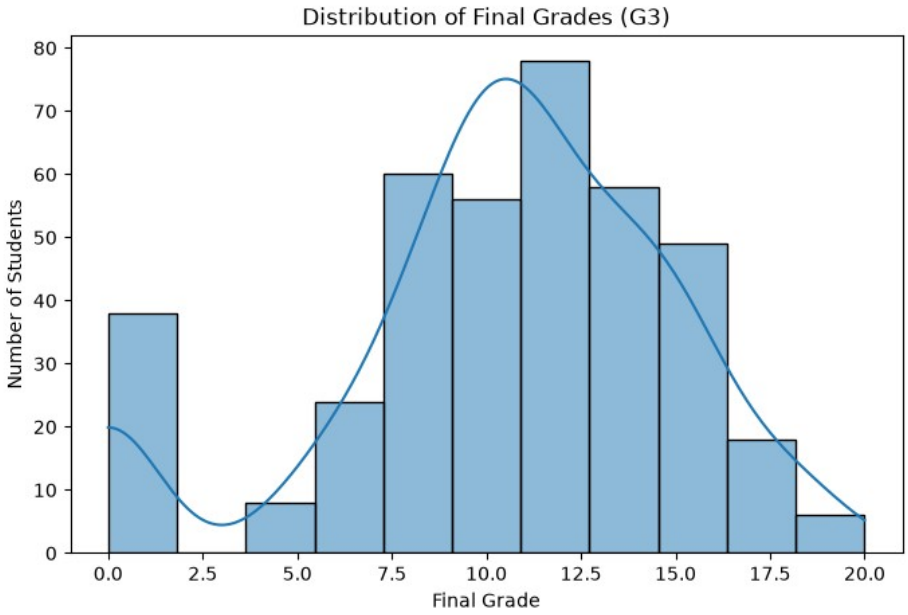



**Figure 1: Distribution of Final Grades (G3)**

The distribution shows how final grades vary across students and provides a view of the most common grade ranges and the overall spread.

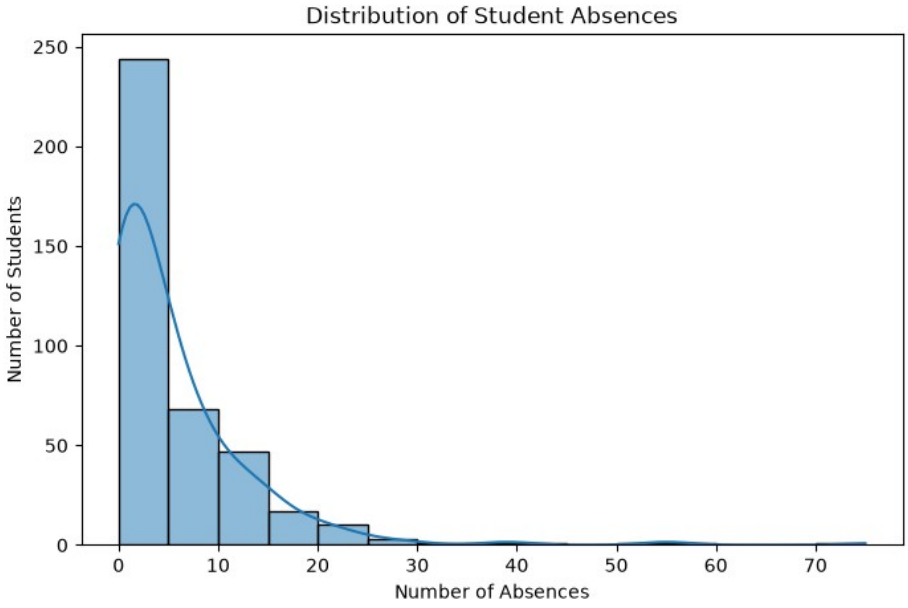


**Figure 2: Distribution of Student Absences**

The distribution shows how frequently different absence levels occur and highlights the presence of students with comparatively high absence counts.

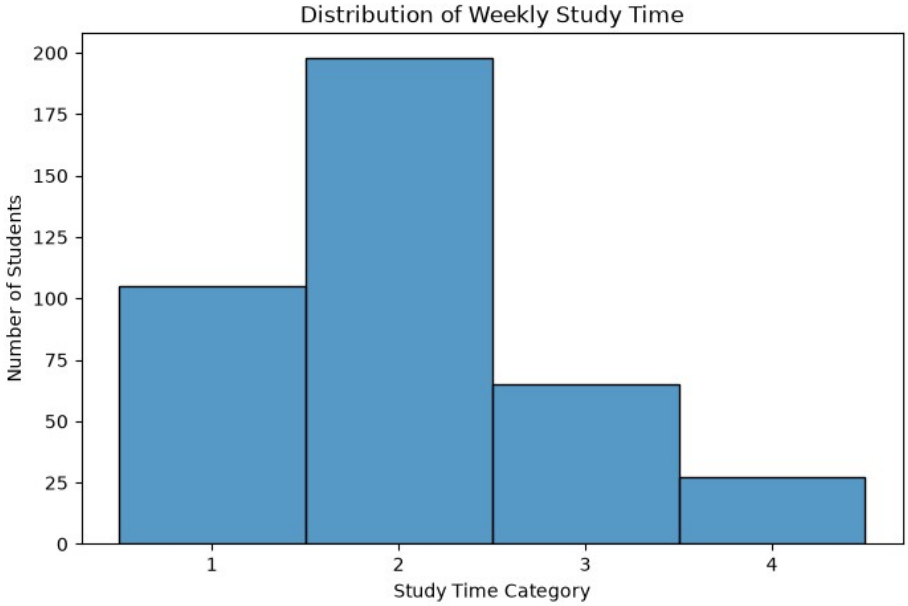



**Figure 3: Distribution of Weekly Study Time**

The distribution shows how students are spread across the four weekly study-time categories.

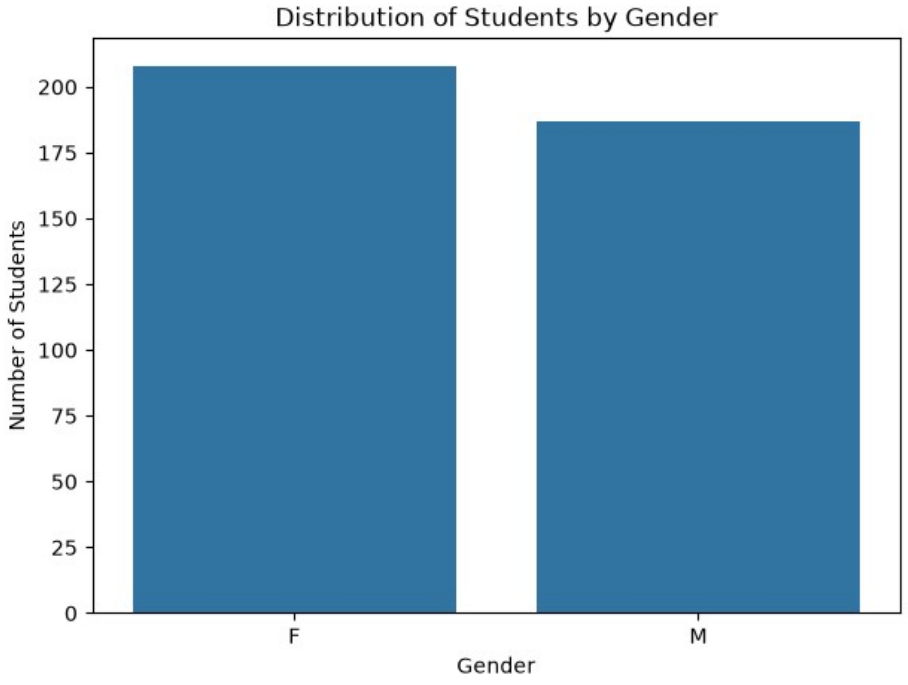



**Figure 4: Distribution of Students by Gender**

The count plot shows the composition of the dataset across the two recorded gender categories.

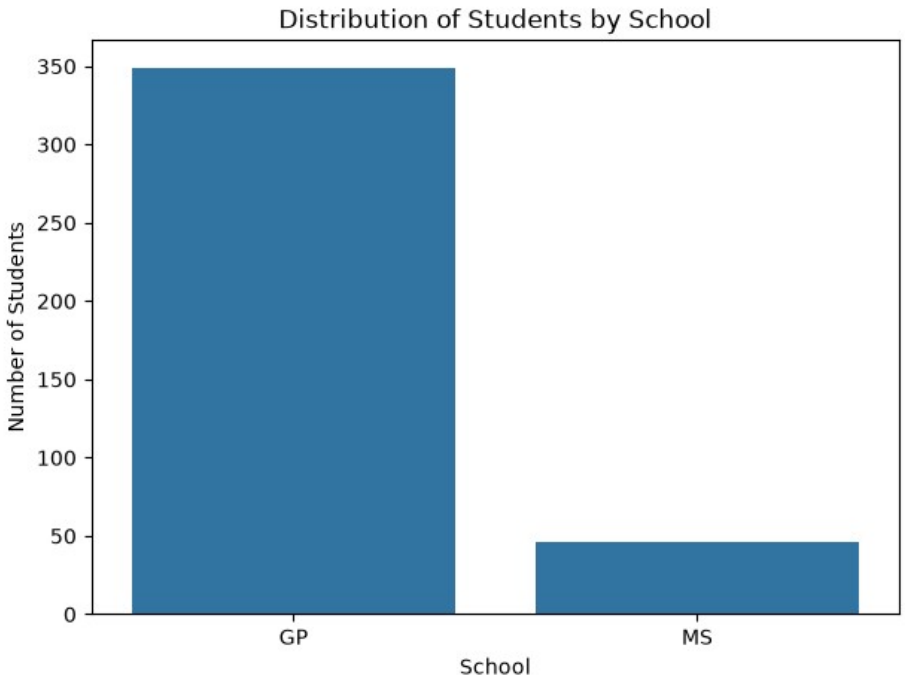



**Figure 5: Distribution of Students by School**

The plot shows the number of students represented from each of the two schools in the dataset.

## 5.2 Bivariate Analysis

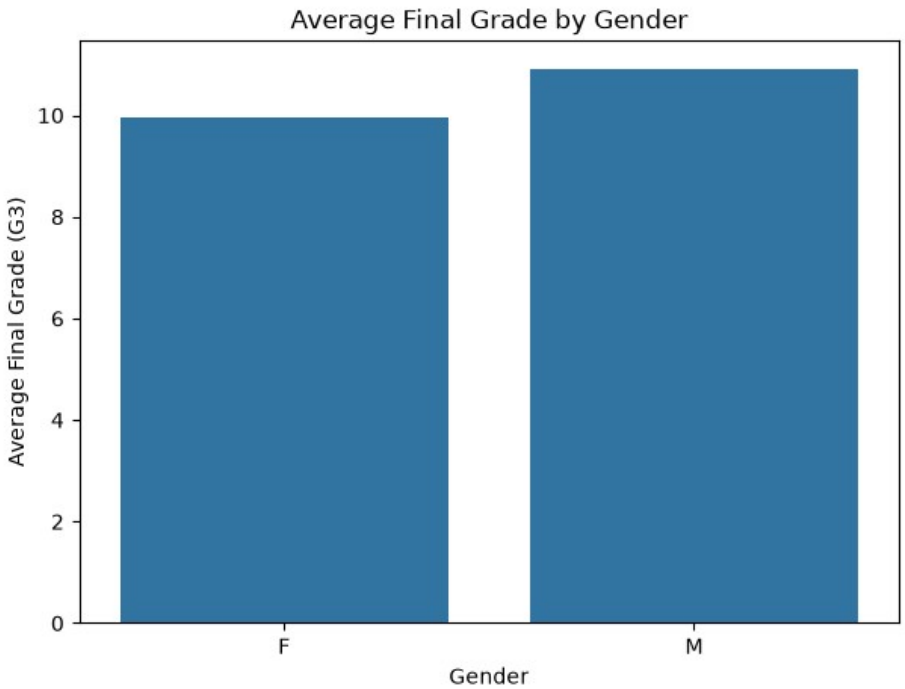


**Figure 6: Average Final Grade by Gender**

The bar chart provides a descriptive comparison of average final grade between the two gender groups.

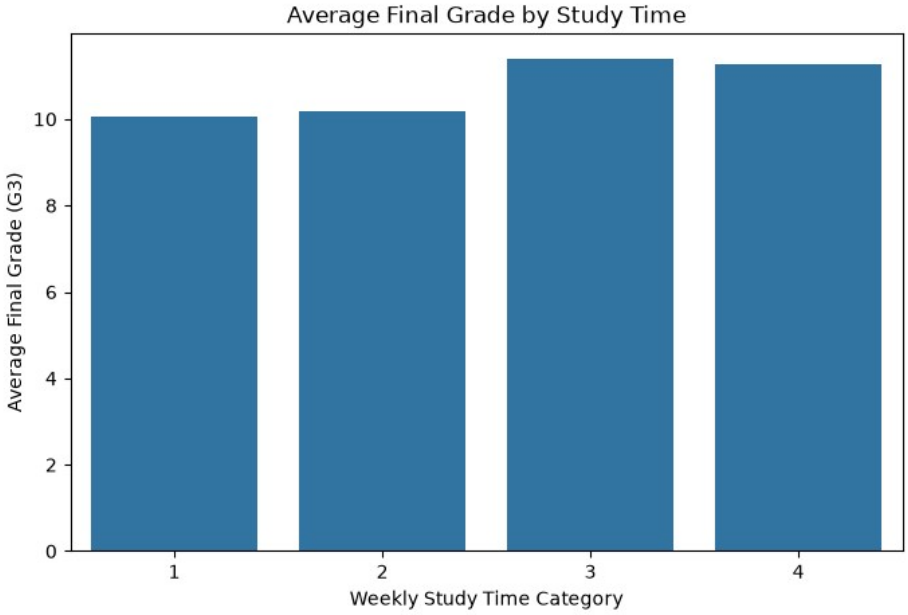



**Figure 7: Average Final Grade by Study Time**

The chart compares average final grades across weekly study-time categories and provides a descriptive view of their association with performance.

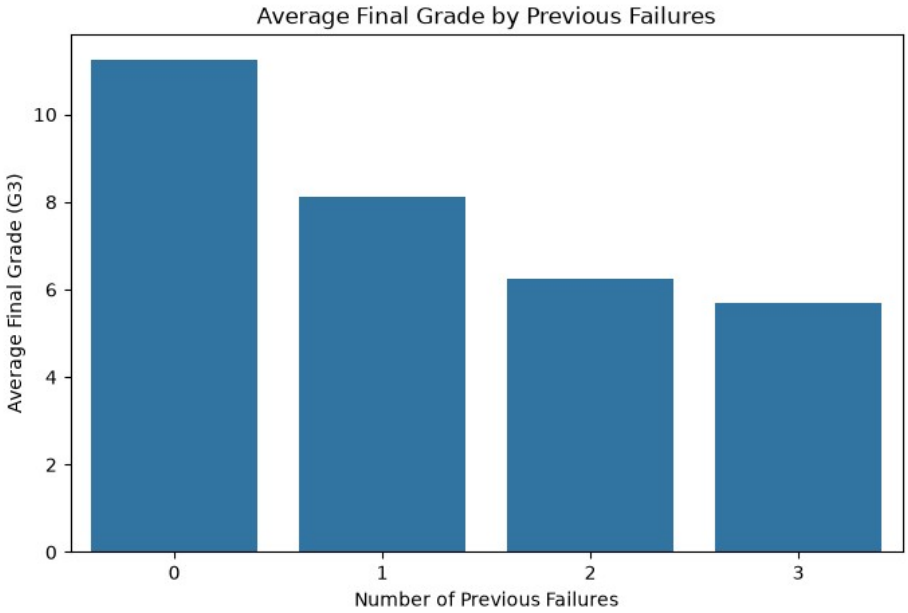


**Figure 8: Average Final Grade by Previous Failures**

The chart compares average final grades across groups defined by the number of previous class failures.

## 5.3 Box Plot Analysis

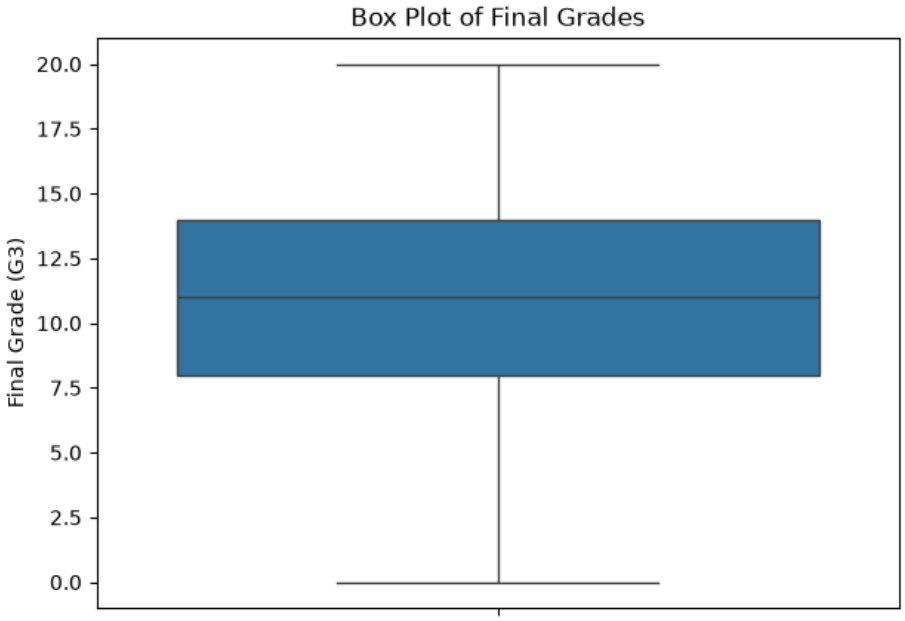



**Figure 9: Box Plot of Final Grades**

The box plot summarizes the central tendency and spread of G3 and helps identify potential extreme observations.

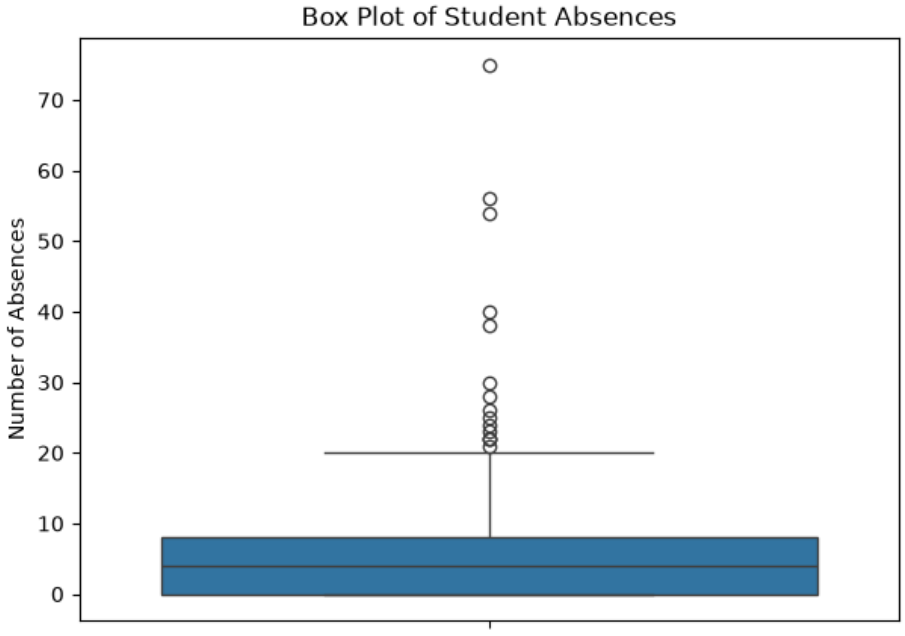


**Figure 10: Box Plot of Student Absences**

The box plot summarizes the spread of absences and highlights observations with comparatively high absence counts.

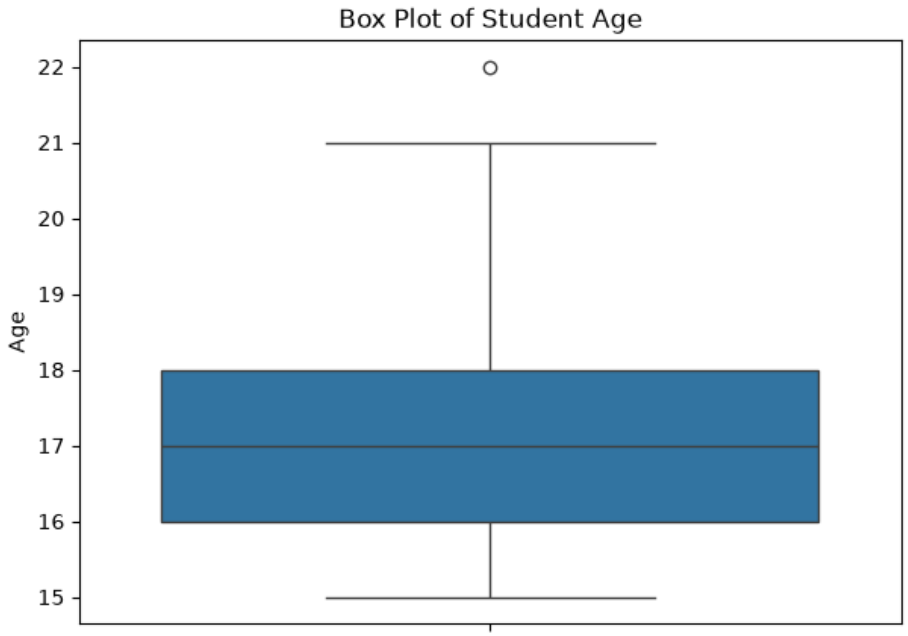



**Figure 11: Box Plot of Student Age**

The box plot summarizes the student age distribution and helps identify a potential age-related extreme observation.

## 5.4 Relationship Analysis

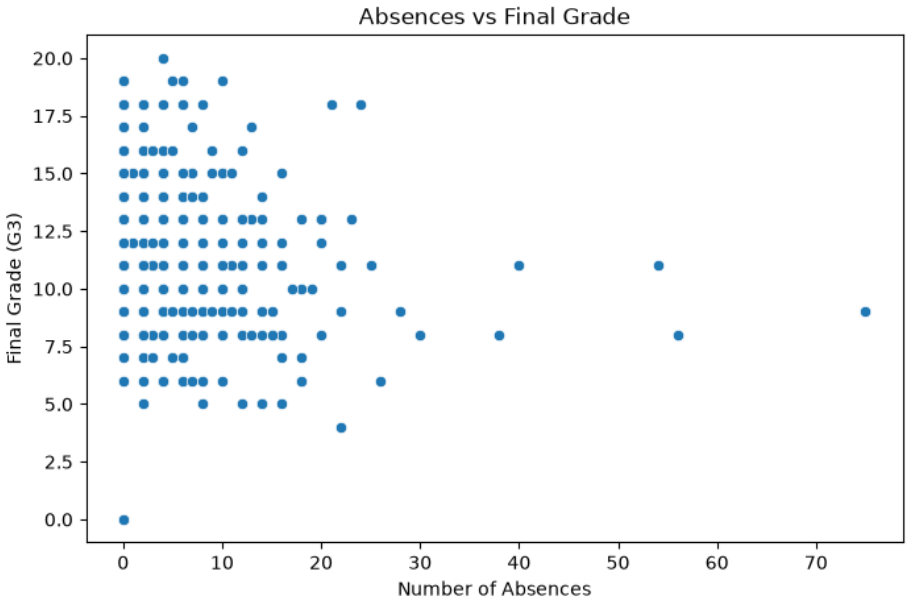


**Figure 12: Absences vs Final Grade**

The scatter plot examines the relationship between absences and final grade without treating the observed association as causal.

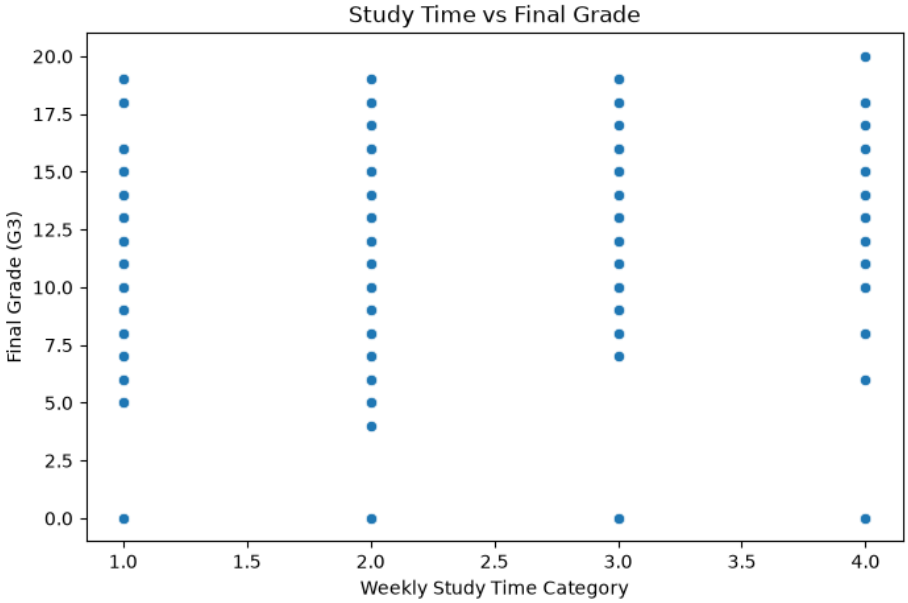



**Figure 13: Study Time vs Final Grade**

The scatter plot examines how final grades are distributed across weekly study-time categories.

## 5.5 Correlation Analysis

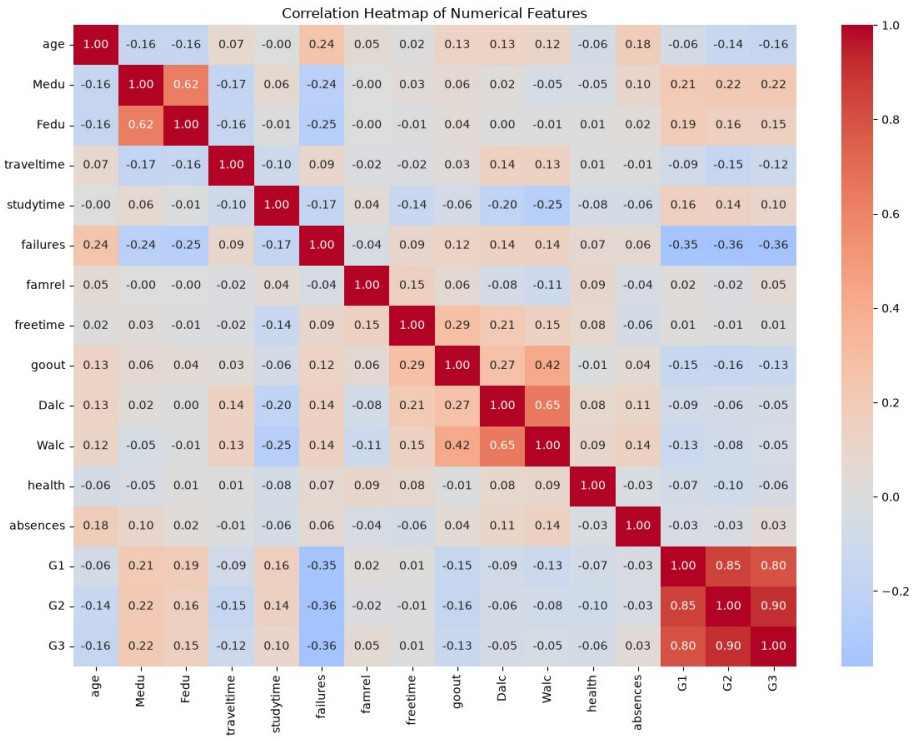



**Figure 14: Correlation Heatmap of Numerical Features**

The heatmap shows the direction and strength of pairwise linear relationships among the numerical variables.

The notebook's correlation-with-G3 ranking is reproduced below. The coefficients describe linear association; they do not establish causation.

| Numerical feature | Correlation with G3 |
|---|---|
| G3 | 1.000000 |
| G2 | 0.904868 |
| G1 | 0.801468 |
| Medu | 0.217147 |
| Fedu | 0.152457 |
| studytime | 0.097820 |
| famrel | 0.051363 |
| absences | 0.034247 |
| freetime | 0.011307 |
| Walc | -0.051939 |
| Dalc | -0.054660 |
| health | -0.061335 |
| traveltime | -0.117142 |
| goout | -0.132791 |
| age | -0.161579 |
| failures | -0.360415 |

Within the notebook's numerical correlation ranking, G2 and G1 have the strongest positive correlations with G3, while failures has the strongest negative correlation among the listed predictors. Medu and Fedu also show positive correlations. The remaining variables have smaller-magnitude linear correlations with G3.

## 5.6 Additional Analysis

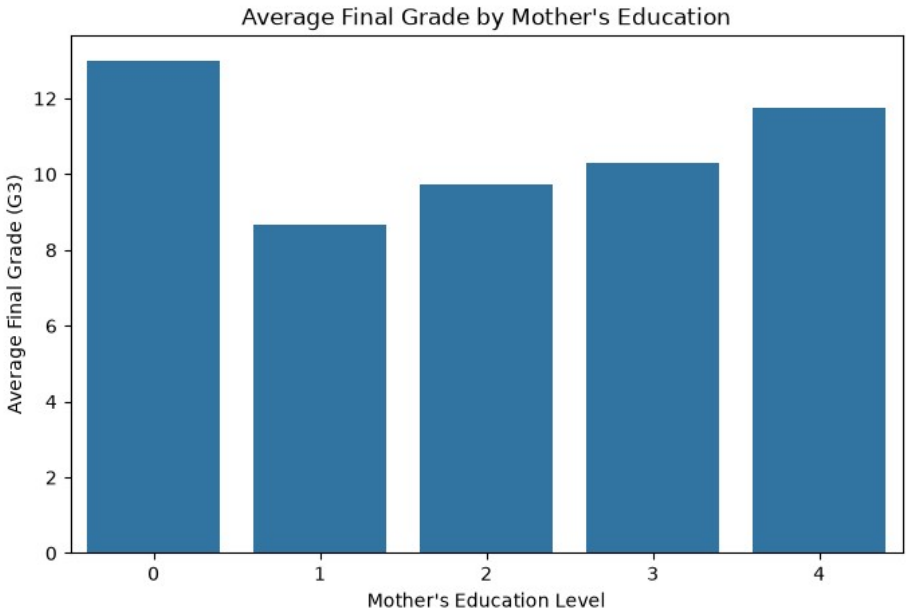



**Figure 15: Average Final Grade by Mother's Education**

The chart compares average final grades across levels of mother's education and provides a descriptive view of the observed association.

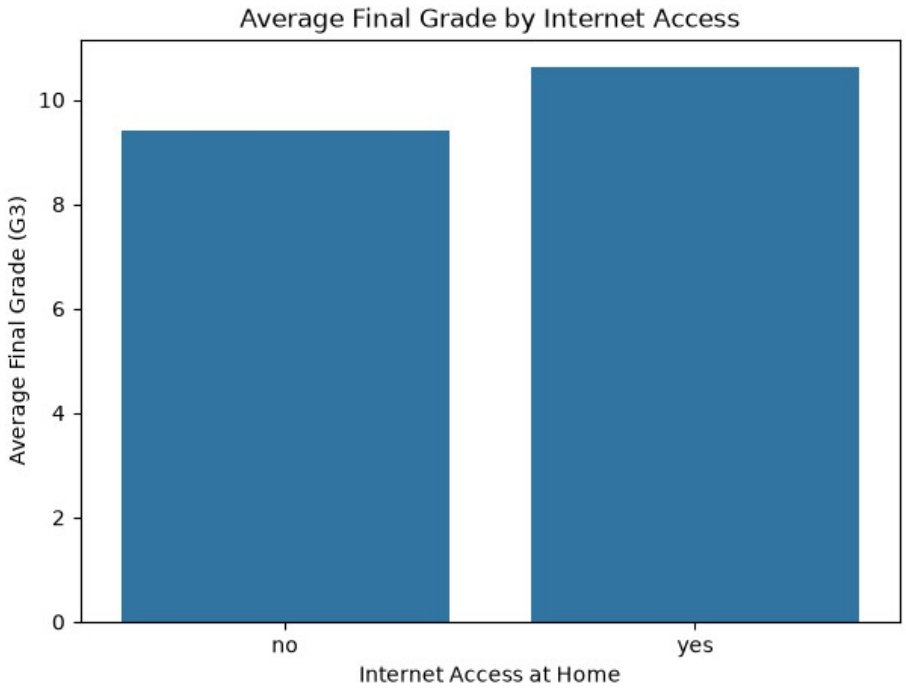



**Figure 16: Average Final Grade by Internet Access**

The chart compares average final grades for students with and without internet access at home.

These additional visualizations extend the descriptive analysis to parental education and internet access, both of which are included in the notebook's EDA.

# 6. Key Findings & Insights

**Final Grade Distribution:** The G3 distribution describes variation in final academic performance across the 395 students.

**Study Time:** Students are distributed across four study-time categories, and the notebook compares average G3 across these categories.

**Previous Failures:** Average final grades vary across groups with different numbers of previous failures, indicating an observed association between prior academic history and final performance.

**Absences:** Absence levels vary across students, with the IQR analysis identifying 15 potential outliers and the scatter plot examining their relationship with G3.

**Gender:** Both gender categories are represented, and their average final grades are compared descriptively.

**Parental Education:** Average G3 is compared across levels of mother's education; Medu has a positive numerical correlation with G3 of 0.217147.

**Internet Access:** The notebook compares average G3 for students with and without internet access at home.

**Correlation:** G2 (0.904868) and G1 (0.801468) have the strongest positive linear correlations with G3 in the notebook's ranking, while failures has the strongest negative correlation at -0.360415 among the listed variables.

**Outliers:** Potential outliers were identified for age, absences, and G2 using the IQR method; no potential outliers were reported for G1 or G3. The notebook does not remove them.

These findings are descriptive associations observed in the dataset. They should not be interpreted as evidence that a factor directly causes a change in academic performance.

# 7. Statistical Summary

The statistical summary provides a compact reference for the scale of the dataset and the observed final-grade and absence measures. It complements the visual analysis by presenting the principal student-performance indicators in a single table.

| Measure | Value |
|---|---|
| Total students | 395 |
| Average Final Grade (G3) | 10.42 |
| Median Final Grade (G3) | 11.0 |
| Highest Final Grade | 20 |
| Lowest Final Grade | 0 |
| Average Absences | 5.71 |

The values above are the exact summary values printed by the notebook, with the average values rounded as displayed there.

# 8. BLEEP EDA Requirement Mapping

| Requirement | Status | Evidence in Project |
|---|---|---|
| Dataset understanding | Completed | Dataset loaded and examined through structure, dimensions, types, and descriptive statistics. |
| Dataset dimensions | Completed | 395 rows and 33 columns. |
| Data types | Completed | 16 numerical and 17 categorical features. |
| Descriptive statistics | Completed | Numerical and categorical summaries included. |
| Data dictionary | Completed | Important project variables documented. |
| Missing value analysis | Completed | 0 missing values. |
| Duplicate analysis | Completed | 0 duplicate rows. |
| Validity checks | Completed | Observed ranges and category values checked. |
| Outlier analysis | Completed | IQR-based checks for age, absences, G1, G2, G3. |
| Univariate analysis | Completed | Grade, absences, study time, gender, and school distributions. |
| Bivariate analysis | Completed | Group comparisons and pairwise visualizations. |
| Multivariate analysis | Completed | Numerical correlation matrix/heatmap. |
| Numerical analysis | Completed | Descriptive statistics and correlation analysis. |
| Categorical analysis | Completed | Frequency and group-wise analyses. |
| Correlation analysis | Completed | Correlation matrix, heatmap, and G3 ranking. |
| Histograms | Completed | 3 histograms in notebook. |
| Box plots | Completed | 3 box plots in notebook. |
| Bar charts | Completed | 3 bar charts in notebook. |
| Count plots | Completed | 2 count plots in notebook. |
| Scatter plots | Completed | 2 scatter plots in notebook. |
| Correlation heatmap | Completed | Numerical-feature heatmap included. |
| Additional visualizations | Completed | Mother's education and internet access vs G3. |
| Visualization interpretations | Completed | Notebook includes interpretation text for each visualization. |
| Key findings | Completed | Dedicated findings section in notebook/report. |

| Requirement | Status | Evidence in Project |
|---|---|---|
| Conclusion | Completed | Notebook includes conclusion and next steps. |
| Future scope | Completed | Further statistical analysis, prediction, evaluation, and explainability suggested. |
| Time-series analysis | Not Applicable | The dataset does not contain a suitable time-based variable. |

# 9. Project Workflow

Each stage in the workflow contributes to the final interpretation: the early stages establish what the data contain and whether they are suitable for analysis, while the later stages examine distributions, relationships, correlations, and visual evidence. The workflow therefore provides a traceable path from the raw dataset to the reported findings.

The completed EDA follows the workflow below.

**Dataset**

↓

**Data Loading**

↓

**Data Understanding**

↓

**Feature Classification**

↓

**Data Quality Checks**

↓

**Validity & Consistency Checks**

↓

**Outlier Detection**

↓

**Univariate Analysis**

↓

**Bivariate Analysis**

↓

**Multivariate Analysis**

↓

**Correlation Analysis**

↓

**Visualization & Interpretation**

↓

**Key Findings**

↓

**Conclusion**

## 10. AI / GenAI Usage

AI/GenAI support is treated as an assistance layer rather than as a substitute for the completed analytical workflow. The reported statistics, visualizations, and interpretations remain those produced and documented in the supplied notebook.

The completed notebook documents a Python/Jupyter-based exploratory workflow using pandas, NumPy, Matplotlib, and Seaborn. It does not document AI/GenAI being used to perform the dataset analysis or generate the reported results. Accordingly, this report does not attribute any analytical result to an AI/GenAI system.

## 11. Conclusion

This exploratory data analysis examined the student-mat dataset to understand demographic, academic, family, social, behavioral, and study-related characteristics in relation to final academic performance represented by G3. The completed notebook covers dataset understanding, data-quality checks, univariate analysis, bivariate analysis, multivariate/correlation analysis, and outlier detection.

The visualizations provide descriptive views of grade distributions, absences, study-time categories, gender and school composition, group-wise final grades, relationships involving absences and study time, and additional comparisons involving mother's education and internet access. The correlation analysis provides a focused view of linear relationships between numerical variables and G3.

Overall, the EDA provides a structured understanding of the available student-performance data and establishes a foundation for future statistical analysis or predictive work.

## 12. Limitations

- The analysis is observational and descriptive; it does not establish causation.
- The findings are specific to the supplied student-performance dataset and its available variables.
- The dataset does not contain a suitable time-based variable, so time-series analysis is not applicable.
- The scope is limited to exploratory analysis and does not include a completed predictive model or model evaluation.

## 13. Future Scope

The following are future extensions, not completed components of this project:

- Perform deeper statistical analysis of observed relationships.
- Investigate and engineer potentially useful features.
- Apply suitable predictive modeling techniques for student performance.
- Compare multiple predictive models using appropriate evaluation metrics.
- Explore explainable-AI techniques to interpret future model predictions.

# 14. References

1. student-m.csv — project dataset used in the completed notebook.

2. Student Performance & Academic Analytics — completed Jupyter Notebook supplied for this BLEEP internship project.

No additional external analytical references are claimed in the completed notebook.

# 15. Appendix

### A. Complete Numerical Feature List

age, Medu, Fedu, traveltime, studytime, failures, famrel, freetime, goout, Dalc, Walc, health, absences, G1, G2, G3.

### B. Complete Categorical Feature List

school, sex, address, famsize, Pstatus, Mjob, Fjob, reason, guardian, schoolsup, famsup, paid, activities, nursery, higher, internet, romantic.

### C. EDA Checklist

✓ 3 histograms

✓ 3 box plots

✓ 3 bar charts

✓ 2 count plots

✓ 2 scatter plots

✓ 1 correlation heatmap

✓ 2 additional meaningful visualizations

✓ Interpretation of visualizations

✓ Key findings and insights

✓ Conclusion and next steps

Time-series analysis: Not Applicable — the dataset does not contain a suitable time-based variable.